

Deep Transfer Learning for Face Presentation Attack Detection Using ResNet-18 and VGG-16

Maahum Khan

Master of Computer Engineering

Queen's University

Kingston, ON, Canada

20mk@queensu.ca

Abstract—As facial recognition technology becomes increasingly integrated into sensitive digital authentication systems, safeguarding against presentation attacks such as printed photos and digital screen replays remains a critical security challenge. This paper presents a comparative evaluation of deep transfer learning for face anti-spoofing using the LCC-FASD dataset, investigating the trade-offs between residual learning backbones (ResNet) and deep sequential architectures (VGG). Through structured data preprocessing, augmentation, and iterative hyperparameter optimization, I implemented and fine-tuned both models to handle severe class imbalance and unconstrained real-world variability. The experimental results demonstrate that the optimal ResNet-18 model achieved an exceptional overall test accuracy of 96.95% and a weighted F1-score of 0.9718, successfully isolating fine-grained liveness textures while avoiding the minority-class collapse observed in the sequential VGG-16 architecture.

Index Terms—deep learning, face anti-spoofing, computer vision, presentation attack, ResNet, VGG, biometric security, liveness detection.

I. INTRODUCTION

As the world becomes more digitized and technology is further integrated in every area of human life, the need for reliable digital authentication systems quickly grows. Facial recognition, in particular, has woven into the fabric of everyday technology, including in highly sensitive applications. From unlocking smartphones to verifying mobile banking transactions and passing through automated airport gates, our faces have become one of our primary digital keys. Due to the global demand for frictionless, contactless authentication, the biometric security sector is experiencing immense financial growth. Recent analysis of the global facial recognition market revealed that it is expected to grow from a value of approximately USD\$8.58 billion in 2025 to an estimated USD\$20.88 billion by 2031 [1].

However, this widespread adoption exposes a critical blind spot: standard facial recognition systems are designed to verify *who* the presented face is, not *if* that face is physically present. This leaves these systems vulnerable to presentation attacks where impostors bypass security using simple, inexpensive mediums such as high-resolution printed photos, looping videos on a smartphone screen, or 3D silicone masks as exemplified in Figure 1 [2]. Because these exploits can lead to severe financial fraud and security breaches, the implemen-

tation of reliable face anti-spoofing countermeasures is now an essential requirement for secure biometric infrastructure.

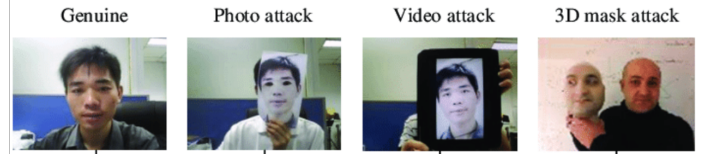


Fig. 1. Example of potential face spoofing attacks.

Distinguishing between a live human subject and a presentation artifact is a distinct technical challenge in computer vision. Unlike standard facial recognition which seeks invariant features to map human identity across varying poses and illuminations, face anti-spoofing requires a model to detect subtle, high-frequency artifacts unique to spoof mediums. These artifacts include microscopic print textures, moiré patterns generated by digital screen captures, loss of depth cues, and abnormal specular reflections. Early computer vision techniques heavily relied on hand-crafted feature descriptors such as Local Binary Patterns (LBP) [3] [4] and Histograms of Oriented Gradients (HOG) [5] [4], paired with classical classifiers. Although these hand-crafted methods were computationally lightweight, they consistently struggle in unconstrained real-world conditions, and often fail to generalize across diverse lighting environments, camera sensors, and attack variations. This limitation has led researchers to turn to deep learning to tackle the task. Deep Convolutional Neural Networks (CNNs) automatically learn hierarchical representations that are capable of capturing complex spatial and textural anomalies.

However, selecting and optimizing the right neural network architecture for the task is a central challenge, particularly when balancing model complexity and high representational capacity with inference latency and generalization risks like overfitting. This paper investigates the effectiveness of deep CNN architectures for presentation attack detection using the LCC-FASD dataset [6], which contains heavy class-imbalance (mostly contains spoof samples which is inherent to real-world biometric security scenarios). Specifically, I fine-tune and conduct a rigorous comparative evaluation of two major computer vision models: ResNet-18, which incorporates skip

connections to facilitate deep gradient flow and isolate fine-grained liveness textures, and VGG-16, a deep sequential convolutional network.

The primary contributions of this work are threefold:

- 1) I develop a face anti-spoofing detection model with a peak accuracy of 96.95% and weighted F1-score of 0.9718 with minimal inference latency and computational overhead unlike larger, more complex models.
- 2) I design, train, and evaluate two distinct architectures, ResNet-18 and VGG-16, side-by-side to explore their capability in detecting liveness artifacts and how they respond when there is class imbalance.
- 3) I establish a complete, reproducible training framework that incorporates data augmentation, weight decay regularization, and structured unfreezing strategies to mitigate the severe overfitting tendencies that are common to deeper sequential vision models.

II. RELATED WORK

Developing an effective automated Face Anti-Spoofing (FAS) system requires an understanding of how foundational computer vision models evolved from hand-crafted texture descriptors to deep convolutional neural networks (CNNs). This section examines five key studies that trace this architectural evolution, concluding with a comparative analysis of the chosen baseline and state-of-the-art models.

A. Foundational Computer Vision Models

Many of the foundational models in computer vision emerged from the Image Large-Scale Visual Recognition Challenge (ILSVRC) in the early 2010s, where researchers sought to develop models that classify images in the ImageNet dataset (with 1.28 million training images) into 1000 classes. Two of these foundational models are used as feature representation learning backbones in this project: ResNet and VGG. Prior to 2014, existing CNNs used few, large convolutions or filters to extract features and learn from visual input. To improve feature representation for large-scale visual recognition, Simonyan and Zisserman (2014) investigated the impact of network depth in their VGG architecture. By uniformly stacking small 3x3 convolutional filters instead of using large, heterogeneous receptive fields, VGG demonstrated that increasing network depth significantly boosts accuracy, achieving an 8.1% top-5 error rate on the ImageNet ILSVRC dataset. However, VGG's sequential design relies heavily on dense fully connected layers, resulting in approximately 138 million parameters, high memory footprints, and slow inference speeds [7].

Simultaneously, Szegedy et al. (2014) introduced GoogLeNet, addressing multi-scale feature learning through parallel processing branches known as Inception blocks. By allowing the network to evaluate 1x1, 3x3, and 5x5 convolutions simultaneously, GoogLeNet won the ILSVRC 2014 classification challenge with a 7.9% top-5 error rate. Despite its high efficiency, the multi-branch structure introduces architectural complexity that makes direct gradient flow optimization difficult [8].

GoogLeNet, alongside many deep networks, demonstrated how high depth can lead to vanishing gradients, where the change applied to perceptrons very deep in a network can become extremely small and prevent the perceptron from learning. To resolve this, He et al. (2015) proposed ResNet, a CNN that incorporated residual skip connection that perform identity mapping. These skip connections allowed gradients to flow uninterrupted across layers which enabled ultra-deep networks (such as ResNet-50 and ResNet-101, the 50-layer and 101-layer versions of ResNet) to win ILSVRC 2015 with a remarkable 3.57% top-5 classification error rate on the ImageNet dataset. In He et al.'s original benchmark, ResNet was directly evaluated against VGG-16/19 backbones, demonstrating superior accuracy and faster convergence by overcoming VGG's depth bottlenecks using its skip connection mechanism [9].

B. Face Anti-Spoofing Detection Models

Prior to 2014, solutions to face anti-spoofing detection tasks relied on classical machine learning techniques that manually extracted features. The transition from shallow feature extraction to deep hierarchical representation began when researchers first applied convolutional networks directly to biometric security. Yang et al. (2014) pioneered this shift by introducing a simple 5-layer CNN architecture (made up of three convolutional layers and two fully connected layers) to replace traditional hand-crafted descriptors like Local Binary Patterns (LBP) for face anti-spoofing on the CASIA-FASD and Replay-Attack datasets [4]. While this foundational work proved that end-to-end neural feature learning was a viable solution, the shallow design suffered from a high half-total error rate ranging between 10% and 16%, and struggled with cross-dataset generalization [10].

While Yang et al.'s simple architecture provided the useful benefit of little to no inference latency in real-world applications, it lacked in accuracy and performance. In contrast, Liu et al. (2018) addressed specialized presentation attack detection with a solution that achieved a near-perfect accuracy. They developed a dual-stream CNN-RNN architecture that utilized auxiliary supervision (combining binary classification with depth map and reflection supervision. Tested across the OULU-NPU, CASIA-MSFD, and MSU-MSFD datasets, this model achieved state-of-the-art results with an average classification error rate near 1%. Nevertheless, this heavy dual-stream design incurs massive computational overhead and high inference latency, making it impractical for resource-constrained or real-time deployment scenarios [11].

C. Comparison of Related Literature

Table I compares the five models in the literature reviewed.

III. METHODOLOGY

Building a reliable face anti-spoofing (FAS) system requires an end-to-end pipeline capable of ingesting raw images, normalizing visual characteristics, extracting high-frequency textural patterns, and executing binary classification. This

TABLE I
COMPARISON TABLE OF FIVE RELATED LITERATURE PAPERS

Model	Application	Dataset Used	Method	Results	Limitations
VGG [7]	Image classification	ImageNet	Stacking many uniform 3x3 convolutional layers	Strong general feature extractor. Top-5 error: 8.1%	Heavy parameter count, slow training
GoogLeNet [8]	Image classification	ImageNet	Inception blocks, multi-scale parallel processing	Strong feature extraction. Top-5 error: 7.9%	Complex architecture, computationally expensive, inference latency
ResNet [9]	Image classification	ImageNet & CIFAR-10	Skip connections (residual learning)	No vanishing gradient problem, more efficient model. Top-5 error rate: 3.7%	Identity mapping assumptions, high memory footprints
CNN for FAS [10]	Face anti-spoofing	CASIA-FASD & Replay-Attack	5-layer CNN (3 convolutional, 2 FC layers)	Half-Total Error Rate: 10-16%	Poor cross-dataset generalization, low baseline accuracy, too simple for application
Auxiliary Supervised CNN-RNN [11]	Face anti-spoofing	OULU-NPU, CASIA-MFSD, & MSU-MSFD	Dual-stream CNN-RNN architecture with auxiliary supervision	Avg. Classification Error Rate: 1%	Extremely heavy computation load and high inference latency

framework utilizes the Large Crowd-Collected Face Anti-Spoofing Dataset (LCC-FASD) to train and evaluate both architectures.

A. End-to-End System Overview and Data Flow

The operational workflow follows a sequential pipeline from input data ingestion to the final liveness detection. Raw video frames or face images are loaded via dataset split lists, passing first through automated preprocessing and tensor resizing to standardize spatial dimensions.

Once it's normalized, the image tensors enter the chosen convolutional neural network backbone, either ResNet-18 or VGG-16, to extract spatial and textural features. The final fully connected layers compress these feature maps into a binary classification logit, outputting a clear determination of whether the subject is genuine or an artificial presentation attack. The end-to-end system data flow is summarized in Figure 2 below.

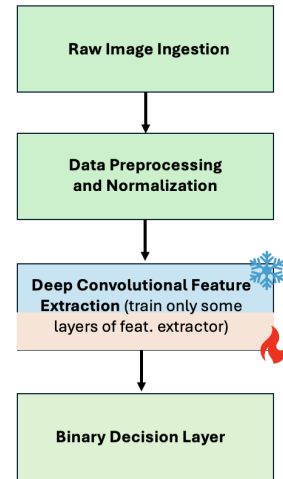


Fig. 2. End-to-end system data flow diagram.

B. Dataset Description and Challenges

The project uses the LCC-FASD dataset (Large Crowd-Collected Face Anti-Spoofing Dataset), selected specifically because it moves past the limitations of older, studio-controlled

benchmarks by gathering unconstrained imagery across diverse real-world environments. Additionally, the dataset is majority spoof cases, which allows us to test each model's ability to understand and detect artifacts of presentation attacks, and determine how each approach (residual learning vs. deep, sequential architecture) responds to class imbalance.

LCC-FASD contains 18.8 thousand samples including genuine human faces and various presentation attack types, including high-resolution printed photos and digital screen replays captured across different devices [6]. Prior to using LCC-FASD, several different datasets containing fewer samples were tried and tested, but they lacked variety in lighting and artifacts, making them less generalizable and robust for real-world applications, and their small size ultimately led to the models easily overfitting.

The primary challenge is in unconstrained real-world variability, such as changing ambient illumination, compression artifacts, and varied camera angles. Because spoof indicators (like print dot patterns or screen moiré effects) are subtle, the model must isolate high-frequency spatial anomalies rather than relying on facial identity features.

C. Model Architectures and Component Design

To evaluate how structural mechanics influence liveness detection, two distinct convolutional architectures were implemented under identical experimental conditions: VGG-16 and ResNet-18. Unlike heavy multi-stream architectures (e.g., Liu et al., 2018 [11]), massive transformer networks, or very simple architectures (e.g., Yang et al., 2014 [10]), VGG-16 and ResNet-18 offer an optimal balance of feature representation capacity and computational efficiency, making them practical for real-world deployment without excessive inference latency.

VGG-16 relies on a deep sequential stack of uniform 3×3 convolutional filters followed by max-pooling layers, ending in dense fully connected layers. While its uniform filter layout builds strong hierarchical feature maps, its strictly sequential design makes it vulnerable to vanishing gradients, which can hinder the extraction of delicate liveness textures [7]. The architecture of VGG-16 is seen below in Figure 3. During training, all layers were frozen except for the last block of layers and a fully-connected binary classifier which were trained on the LCC-FASD dataset.

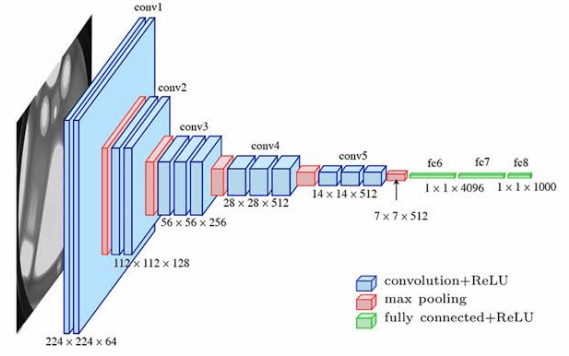


Fig. 3. Architecture of VGG-16 [12].

ResNet18 incorporates residual shortcut connections that allow gradients to bypass one or more layers, enabling direct identity mapping. The residual learning building block is illustrated in Figure 4 from the original 2015 paper on ResNet [9].

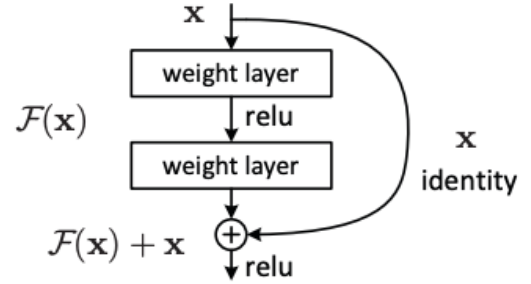


Fig. 4. Architectural diagram of ResNet-18 [13].

This structural improvement prevents training degradation in deeper networks, making it much more effective at capturing fine-grained edge anomalies and surface reflections without losing gradient stability. The overall architecture of ResNet-18 is illustrated in Figure 5 below [13].

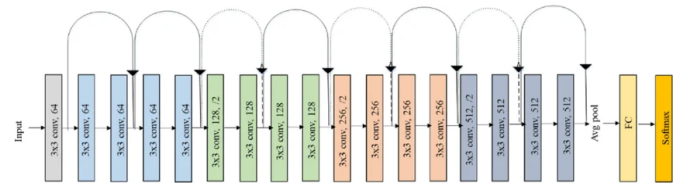


Fig. 5. The architecture of ResNet-18 [13].

IV. IMPLEMENTATION

Carrying out these experiments required establishing a controlled environment, formatting the raw data into consistent tensor shapes, and systematically tuning model behavior through structured freezing strategies, data augmentations, and hyperparameter optimization.

A. Environment and Software Stack

All training routines, data preprocessing scripts, and evaluation loops ran on a dedicated remote server using an NVIDIA GPU to accelerate tensor operations and backpropagation. The codebase was written entirely in Python, utilizing PyTorch and torchvision for model instantiations and deep learning pipelines. Array manipulations and image transformations were handled using NumPy and OpenCV.

B. Data Preprocessing and Augmentation

Because the raw LCC-FASD frames varied widely in resolution, aspect ratio, and lighting, a standardized preparation pipeline was applied:

- 1) **Resizing.** All facial images were scaled to a uniform 224×224 pixel grid to match the input layer dimensions expected by both backbones.
- 2) **Normalization.** Pixel tensors were normalized using standard channel-wise mean and standard deviation values derived from ImageNet to keep early gradient updates stable.
- 3) **Data Augmentation.** To combat the heavy overfitting tendencies of deep vision models, explicit transformations were integrated into the training loops. These transformations were random resized cropping, rotation, horizontal flipping, colour jitter, erasing, and minor brightness tweaks to force the networks to focus on structural liveness textures rather than background noise. An exemplary sample of a sample image from LCC-FASD with the data augmentations applied is shown in Figure 6.

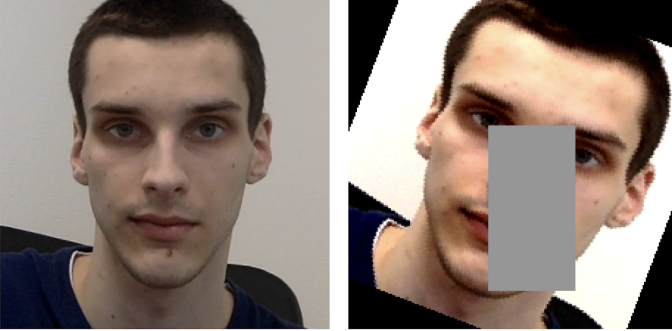


Fig. 6. Side-by-side comparison of a raw (left image) vs. preprocessed data sample (right image) with the data augmentation methods used in the project applied.

C. Data Split Approach

The dataset was partitioned using the protocol-driven text split files provided with LCC-FASD, establishing strict boundaries between training, validation, and test subsets.

This split methodology is optimal because it enforces subject-independent separation. In practical biometric deployment, the system must authenticate faces it has never encountered before. Keeping identities entirely distinct across

the training, validation, and test splits prevents data leakage, forcing the networks to learn generalized spatial-frequency indicators of liveness (such as screen moiré patterns and print dot textures) instead of memorizing specific subjects.

D. Model Training, Optimization, and Iterative Tuning

Training both models required a series of iterative tuning and optimization to find the best balance between model capacity and overfitting risks. Optimization was guided by tracking training loss, validation loss, and validation accuracy, backed by an early stopping checkpointing strategy that saved weights during the epoch yielding the highest validation accuracy. The loss function used was Binary Cross Entropy (BCE) Loss, which was minimized using the following formulation:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (1)$$

In optimization and tuning, various methods were used: freezing and unfreezing of different sections of the backbones, adding or removing dropout, applying or removing weight decay, and tuning hyperparameters such as number of epochs, batch size, learning rate, and weight decay value.

1) *ResNet Optimization and Patterns:* Initial unconstrained training across all layers of ResNet-18 and ResNet-34 caused severe overfitting because the model’s massive capacity quickly outpaced the training signal. Restricting weight updates exclusively to the classification head under-fitted the network, capping validation accuracy near 85%. The optimal balance was achieved by using ResNet-18, selectively unfreezing deeper feature blocks (such as Layer 4), and using a reduced learning rate ($1e - 5$). This configuration allowed the residual architecture to adapt its feature extractors to subtle liveness textures without destabilizing pre-learned spatial priors, resulting in the peak configuration being at a batch size of 32, learning rate of $1e - 5$ with no weight decay, on the 6th epoch. Weight decay regularization was applied and different weight decay rates were tried, but the best results were achieved with this particular configuration which did not apply weight decay. This best configuration achieved a very impressive 94.13% validation accuracy. Conversely, setting learning rates too high ($1e - 4$) caused immediate divergence, while insufficient epochs or excessively constrained layers left the model unable to resolve fine-grained presentation artifacts. The training loss and validation loss graph of the best model is shown in Figure 7, where the final model used the weights saved during the 6th epoch despite 7 epochs having been run.

2) *VGG Optimization and Patterns:* VGG-16 proved highly sensitive to its deep sequential structure and massive parameter count in the fully connected layers (approximately 122M of its 138M parameters). Training only the classifier layer led to an early performance ceiling around 86.25%, while unfreezing higher-level feature blocks triggered immediate overfitting by epoch 2 of training. Overcoming this required only training the last convolution block and last few classifier layers, heavily constraining the learning rate ($1e - 7$) and increasing weight

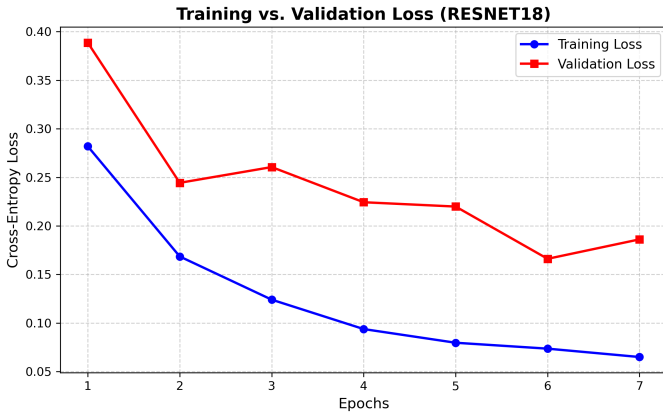


Fig. 7. Training and validation loss of best ResNet model. Final model's weights were saved at Epoch 6, which had the best validation accuracy of 94.13%.

decay ($1e-3$) to inhibit the exploding gradients across the deep sequential stack. Introducing dropout, increasing batch sizes up to 128, or experimenting with different learning rates (and testing all combinations of specifications) consistently caused the model to lock into a persistent performance plateau near 86.26%. Using lower weight decay rates or removing dropout while simultaneously increasing batch size also failed to break the plateau. The consistent plateauing of the model's validation accuracy across many different configurations and combinations is most likely due to its massive size, which makes it very prone to overfitting (at least on the LCC-FASD dataset) and sensitive to even slight changes in hyperparameters. After extensive hyperparameter tuning and model optimization, the optimal balance was achieved with no dropout, a batch size of 32, a learning rate of $1e-7$, weight decay rate of $1e-3$, and running for 4 epochs. This best model achieved a validation accuracy of 86.87%, lower than that of the best ResNet model. Note that this same configuration was tested with higher epochs, but led to validation accuracy falling into the plateau value of 86.26%, so the final model saved was this configuration with 4 epochs. The training and validation loss graph is shown below in Figure 8.

Details of the best VGG and ResNet models are provided in Table II.

V. RESULTS AND DISCUSSION

Related work in face anti-spoofing often develops models that are either too simple with poor performance but no inference latency (i.e. the 5-layer CNN for face anti-spoofing by Yang et al. [10]) or excessively computationally demanding, heavy models with excellent performance but high inference latency (i.e. CNN-RNN auxiliary supervision model by Yang et al. [11]) which makes both options subpar and unrealistic for real-world use due to their insufficiency in either accuracy or inference speed and resource usage. Thus, the main objective of this work was to develop a model that achieves what I consider to be reasonable accuracy for real-world usage of

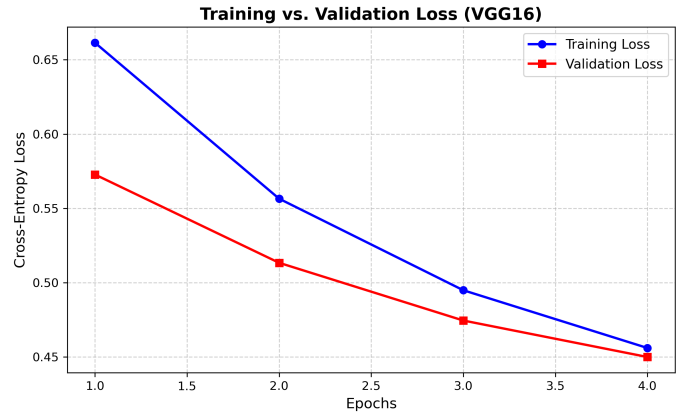


Fig. 8. Training and validation loss of best VGG model.

85% or above without needing to create a large, complex, very computationally demanding model that has notable inference latency like the auxiliary supervised CNN-RNN model by Liu et al. (2018) reviewed in the Related Work. In particular, this project sought to test the hypothesis that convolutional neural networks can effectively learn generalized spatial-frequency indicators of liveness (such as screen moiré patterns, display pixel grids, and print dot textures) to distinguish between genuine facial images and presentation attacks within the LCC-FASD dataset. In an effort to effectively prove this hypothesis, I implemented two versions of the model using two different feature learning backbones (ResNet and VGG) that use different approaches in their architecture and learnability in order to find the best model for this project. I also sought to determine how residual learning architectures (ResNet-18) compare against deep sequential architectures (VGG-16) when handling subject-independent biometric verification and severe class imbalance (since the LCC-FASD dataset contains majority spoof cases). A quantitative summary of the final test evaluation performance of the best ResNet and VGG models is provided in Table III.

A. ResNet-18 Performance Analysis

The test evaluation results strongly validate the hypothesis that residual learning differs in performance from models that rely on deep, sequential network architecture and excellent accuracy can be achieved in this application without requiring extremely large, complex architecture with ResNet-18 achieving an exceptional overall test accuracy of 96.95% and a remarkably low test loss of 0.0849.

1) *Navigating the Capacity-Generalization Tradeoff*: The configuration that succeeded was trained with a batch size of 32, learning rate of $1e-5$, on 7 epochs without weight decay or dropout while selectively training only the last convolution block and fully-connected layers. Early experiments showed that unfreezing all layers caused rapid overfitting because the network's capacity outpaced the training signal, whereas locking everything except the final classifier under-fitted the

TABLE II
SUMMARY OF MODEL SPECIFICATIONS FOR BEST VGG AND RESNET MODELS.

Model Backbone	Portion Trained	Batch Size	Learning Rate	Epochs	Weight Decay	Dropout	Val. Accuracy
ResNet-18	Last conv. block & FC layers	32	$1e - 5$	7	None	None	94.13%
VGG-16	Last conv. block & last few classifier layers	32	$1e - 7$	4	$1e - 3$	$p = 0.5$	86.87%

TABLE III
QUANTITATIVE PERFORMANCE COMPARISON OF RESNET-18 AND VGG-16 ON THE TEST SET OF THE LCC-FASD DATASET. BEST RESULTS IN EACH CATEGORY ARE BOLDED.

Model Backbone	Test Loss	Accuracy	Weighted Precision	Weighted Recall	Weighted F1-Score	Real Class F1-Score
VGG-16	0.3444	95.86%	0.9189	0.9586	0.9383	0.0000
ResNet-18	0.0849	96.95%	0.9756	0.9695	0.9718	0.6899

model. The sweet spot was found by selectively unfreezing Layer 4 alongside the classification head using a moderate learning rate of $1e - 5$. This allowed the residual architecture to adapt its feature extractors to subtle high-frequency liveness textures without breaking the robust spatial priors from the model’s pretraining.

2) *Overcoming Class Imbalance*: The test subset had a heavy class imbalance that is inherent to real-world biometric security scenarios. It contained 7266 spoof samples against only 314 genuine faces. ResNet-18 successfully navigated this skew, securing a spoof class F1-score of 0.98 and a genuine class F1-score of 0.69. This proves that the network successfully learned genuine high-frequency liveness features rather than collapsing into a majority-class predictor. The per-class breakdown of the best ResNet model’s precision, recall, and F1-score are compared in Table IV.

3) *Hyperparameter Sensitivity Insights*: A moderate learning rate of $1e - 5$ successfully avoided the explosive divergence that was observed when training with higher rates ($1e - 4$). Meanwhile, letting the model train for 7 epochs without weight decay provided just enough structural adjustment space to capture fine-grained presentation artifacts. Additionally, using dropout with a probability of 0.5 appeared to help the model’s performance by preventing over-reliance on certain perceptrons and features.

B. VGG-16 Performance Analysis

In contrast, VGG-16 yielded an overall test accuracy of 95.86% with a significantly higher test loss of 0.3444. Despite having an excellent test accuracy, examination of the classification report and per-class metrics revealed critical insights into the structural limitations of deep sequential architectures.

1) *Handling Class Imbalance*: Despite the deceptive 95.86% global accuracy, VGG-16 completely failed to identify genuine faces on the unseen test set, achieving a precision, recall, and F1-score of 0.00 for the real class. Its high global accuracy was entirely an artifact of predicting the majority spoof class with 100% recall. The best VGG-16 model’s

per-class breakdown of precision, recall, and F1-score are summarized in Table V.

2) *Structural Bottlenecks and Gradient Degradation*: VGG-16 has approximately 122M of its total 138M parameters in its massive fully-connected layers. Restricting training solely to the classifier layer while freezing the deep feature stack created an intense representation bottleneck. Because VGG lacks residual skip connections, gradients degraded severely across its deep sequential convolutional blocks. Ultimately, the best performance was achieved by only unfreezing the last few classifier layers and the last convolution block.

3) *The Over-Regularization Trap*: To prevent exploding gradients across the deep sequential stack during training, I tried applying heavy regularization, including high weight decay ($1e - 3$), dropout ($p = 0.5$), and a really small learning rate ($1e - 7$). Although this suppressed training divergence, it heavily over-regularized the network and cut off its capacity to discriminate subtle minority-class features when exposed to unseen test or validation data. Experimenting with higher learning rates ($5e - 6$ to $1e - 3$) or removing weight decay consistently caused the model to lock into a persistent performance plateau near 86.26% validation accuracy. This shows that VGG’s sequential design inherently struggles with fine-tuning of specialized biometric textures.

C. Key Insights and Takeaways

Reflecting on the iterative tuning process and the final test metrics yields three key insights for anti-spoofing system design:

- 1) Global accuracy is a misleading metric in security. In skewed biometric tasks, overall accuracy can mask total failure on critical classes. VGG-16 achieved 95.86% accuracy while completely failing to catch a single genuine user, which shows that monitoring per-class F1-scores, precision-recall tradeoffs, and confusion matrices is crucial in this application.
- 2) Residual shortcuts enable effective transfer learning. Residual paths in ResNet-18 allow gradient adjustments

TABLE IV
QUANTITATIVE PER-CLASS PERFORMANCE OF THE BEST RESNET-18 MODEL ON THE TEST SET OF THE LCC-FASD DATASET.

Class	Precision	Recall	F1-Score
Real	0.5963	0.8185	0.6899
Spoof	0.9920	0.9761	0.9840

TABLE V
QUANTITATIVE PER-CLASS PERFORMANCE OF THE BEST VGG-16 MODEL ON THE TEST SET OF THE LCC-FASD DATASET.

Class	Precision	Recall	F1-Score
Real	0.0000	0.0000	0.0000
Spoof	0.9586	1.0000	0.9788

to smoothly reach deeper feature extractors. This structural advantage is what allowed ResNet-18 to adapt to high-frequency presentation artifacts without sacrificing sensitivity to minority classes.

- 3) While hyperparameter tuning (i.e. learning rate, batch size, epoch count, and weight decay rate adjustments) has a strong impact on a model’s performance, heavy regularization on rigid architectures like VGG-16 can unintentionally strip away the network’s ability to generalize to under-represented real-world classes. Future iterations should explore focal loss formulations or balanced data resampling techniques rather than relying solely on architectural constraints to manage class skew.

VI. CONCLUSION AND FUTURE WORK

This project addressed the vulnerability of facial recognition systems to presentation attacks, such as printed photos and digital screen replays, which bypass standard identity verification. By implementing and evaluating ResNet-18 and VGG-16 on the LCC-FASD dataset, we demonstrated that residual learning backbones can effectively capture subtle, high-frequency liveness textures without destabilizing pre-trained spatial priors. Through structured unfreezing and hyperparameter tuning, our optimal ResNet-18 model achieved a key test accuracy of **96.95%**, proving its capability to distinguish between genuine users and presentation attacks under severe class imbalance.

Future work includes:

- 1) Implementing focal loss, weighted sampling, or synthetic data augmentation to mitigate minority class collapse.
- 2) Testing current model checkpoints on alternative benchmarks like CelebA-Spoof to verify the model’s generalizability across datasets.
- 3) Adapting the architecture into even lighter-weight models for smoother real-time mobile deployment.

REFERENCES

[1] Mordor Intelligence. (2026, June). *Facial recognition market size & share analysis - growth trends and forecast (2026 - 2031)*. <https://mordorintelligence.com>

[2] Sun, W., Song, Y., Zhao, H., & Jin, Z. (2020). A face spoofing detection method based on domain adaptation and lossless size adaptation. *IEEE Access*, 8, 65243–65253.

[3] J. Maatta, A. Hadid, and M. Pietikainen (2011). Face spoofing detection from single images using micro-texture analysis. In *Biometrics (IJCB), 2011 International Joint Conference* (pp. 1–7).

[4] Yan, J., Zhang, Z., Lei, Z., Yi, D., and Li, S. Z. (2012). Face liveness detection by exploring multiple scenic clues. In *Proceedings of the 2012 12th International Conference on Control Automation Robotics & Vision (ICARCV)* (pp. 188–193). IEEE.

[5] J. Komulainen, A. Hadid, and M. Pietikainen (2013). Context based face antispoofing. In *Biometrics: Theory, Applications and Systems (BTAS), 2013 IEEE Sixth International Conference* (pp. 1–8). IEEE.

[6] M. Abbas, M. Munib, M. Sajid, M. Mazher, U. Kalsoom, and L. Chanona-Hernandez (2026). Lightweight Face Anti-Spoofing with MobileNetV2: Transfer Learning, Model Compression, and Evaluation on LCC-FASD. *International Journal of Combinatorial Optimization Problems and Informatics*, 17(4) (pp. 115–133).

[7] K. Simonyan and A. Zisserman (2015). Very deep convolutional networks for large-scale image recognition. In *3rd International Conference on Learning Representations (ICLR)* (pp. 1–14). Computational and Biological Learning Society.

[8] C. Szegedy, W. Liu, Y. Jia, P. Sermanet, S. Reed, D. Anguelov, D. Erhan, V. Vanhoucke, and A. Rabinovich (2015). Going deeper with convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 1–9). IEEE.

[9] K. He, X. Zhang, S. Ren, and J. Sun (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778). IEEE.

[10] J. Yang, Z. Lei, and S. Z. Li (2014). Learn convolutional neural network for face anti-spoofing. In *Biometric Recognition (CCBR 2014)* (Lecture Notes in Computer Science, Vol. 8833, pp. 373–384).

[11] Y. Liu, A. Jourabloo, and X. Liu (2018). Learning deep models for face anti-spoofing: Binary or auxiliary supervision. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 389–398). IEEE.

[12] G. Boesch (2021). Very deep convolutional networks (VGG) essential guide. *Viso.ai*. <https://viso.ai/deep-learning/vgg-very-deep-convolutional-networks/>

[13] F. Ramzan, M. U. G. Khan, A. Rehmat, S. Iqbal, T. Saba, A. Rehman, and Z. Mehmood (2020). A deep learning approach for automated diagnosis and multi-class classification of Alzheimer’s disease stages using resting-state fMRI and residual neural networks. *Journal of Medical Systems*, 44(2), 37.